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Abstract

Student attrition remains a persistent challenge in higher education, with consequences for learners (lost time and debt burden) and institutions (reduced completion rates and inefficient resource planning). This paper studies early prediction of three student outcomes—dropout, continued enrollment, and graduation—using structured institutional data that includes academic progress indicators (e.g., first-semester performance), financial status (e.g., tuition payment and debt), and demographic/background attributes.

We frame the task as a supervised multiclass classification problem and evaluate a baseline pipeline of data preprocessing, feature selection, and comparative modeling using standard machine learning classifiers. Our current results are preliminary: using the course dataset described in our project, we obtain up to 77.9% accuracy on held-out data.

The main contributions are: (i) a problem formulation and feature taxonomy aligned with advising decisions, (ii) an empirical comparison of classical machine learning baselines for early-risk screening, and (iii) a discussion of practical gaps—generalization across cohorts and interpretability—that motivate the next stage of this work.

1 Introduction

Student dropout is a major operational and societal problem for colleges and universities. Institutions seek actionable, early indicators of risk so that advisors can intervene before academic or financial difficulties become irreversible. Prior work has shown that administrative records and early academic signals can support accurate prediction of attrition in higher education [1, 2].

1.1 Problem statement

Given student records containing academic, financial, and background variables observed at enrollment and during the first academic term(s), the goal is to predict a student's eventual outcome: *dropout*, *continued enrollment*, or *graduation*. We target early prediction to support timely intervention.

1.2 Research challenges

Key challenges include: (i) feature availability and timing (many strong predictors appear only after substantial progression), (ii) class imbalance and heterogeneous student populations, and (iii) model generalization across cohorts and programs, where performance can degrade due to dataset shift [5].

1.3 Research objectives and approach

This work aims to (1) identify which academic and financial features are most predictive of risk, and (2) establish transparent baseline models suitable for an early-warning workflow. We implement a standard supervised learning pipeline—data cleaning, train/validation/test splitting, model training, and evaluation—and compare multiple classifiers (e.g., logistic regression and tree-based ensembles), following common practice in student success modeling [3].

1.4 Contributions

The contributions of this paper are:

- A clear early-warning formulation for multiclass outcome prediction (dropout/enrolled/graduated).
- A feature-driven analysis of academic progress versus financial and background variables.
- A baseline experimental benchmark (preliminary) to guide subsequent improvements and interpretability work.

1.5 Paper organization

Section 2 reviews related work on dropout prediction and early-warning systems. Subsequent sections (to be completed) will describe the dataset, methods, experiments, and discussion.

2 Related Work

Research on student outcome prediction spans traditional higher education settings and large-scale online learning environments.

2.1 Dropout prediction in higher education

Aulck et al. model college attrition using large-scale administrative records and show that meaningful prediction is possible even from early transcript information [1]. Lakkaraju et al. propose a machine learning framework aimed at identifying at-risk students sufficiently early to enable intervention, emphasizing operational constraints and the need for actionable models [2]. More recent work has expanded the scope to multiclass success outcomes and public benchmarks;

for example, the "Predicting Student Dropout and Academic Success" dataset and accompanying study provide a widely used reference task for comparing classifiers under a consistent feature set [3].

2.2 MOOC dropout prediction and evaluation pitfalls

In MOOCs, dropout rates are high and clickstream data enable fine-grained behavioral features. Kloft et al. use weekly activity features and support vector machines to predict dropout between course weeks [4]. However, evaluation methodology can lead to overly optimistic estimates when models are trained and tested on the same course; Whitehill et al. highlight this issue and argue for more realistic cross-course evaluation setups [5].

2.3 Gaps and positioning of this work

Across both higher education and MOOCs, two gaps recur: (i) *early* prediction with features available before performance collapse, and (ii) deployment-oriented considerations such as interpretability and stability under cohort shift. Our work positions itself as an early-warning baseline study on structured institutional data, emphasizing feature timing (academic progress vs. financial status) and establishing a reproducible benchmark that can be extended with interpretability analyses and robustness checks.

References

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